LAB 1: INTRODUCTION TO DIGITAL OSCILLOSCOPES

A) Purpose

The main objective of this experiment is to get familiar with the fundamental laboratory equipment. Especially the digital oscilloscope and the signal generator. Moreover, the experiment aims to show the working principles of a breadboard, the importance of probe compensation and the signal measurement skills like triggering and coupling which are important for accurate measurements.

B) Methodology

Task 1: Probe Compensation

First, the oscilloscope probe was connected to Channel 1. To ensure accurate measurements, the probe tip was attached to the oscilloscope's "Probe Comp" terminal, which provides a reference square wave. Initially, the waveform edges were distorted (If the probe is under-compensated or over-compensated, the corners appear rounded or overshoot) and this was where I need to do compensation. After that I adjusted the probe's adjustment screw until the square wave had flat tops which is the sign of a proper compensation.

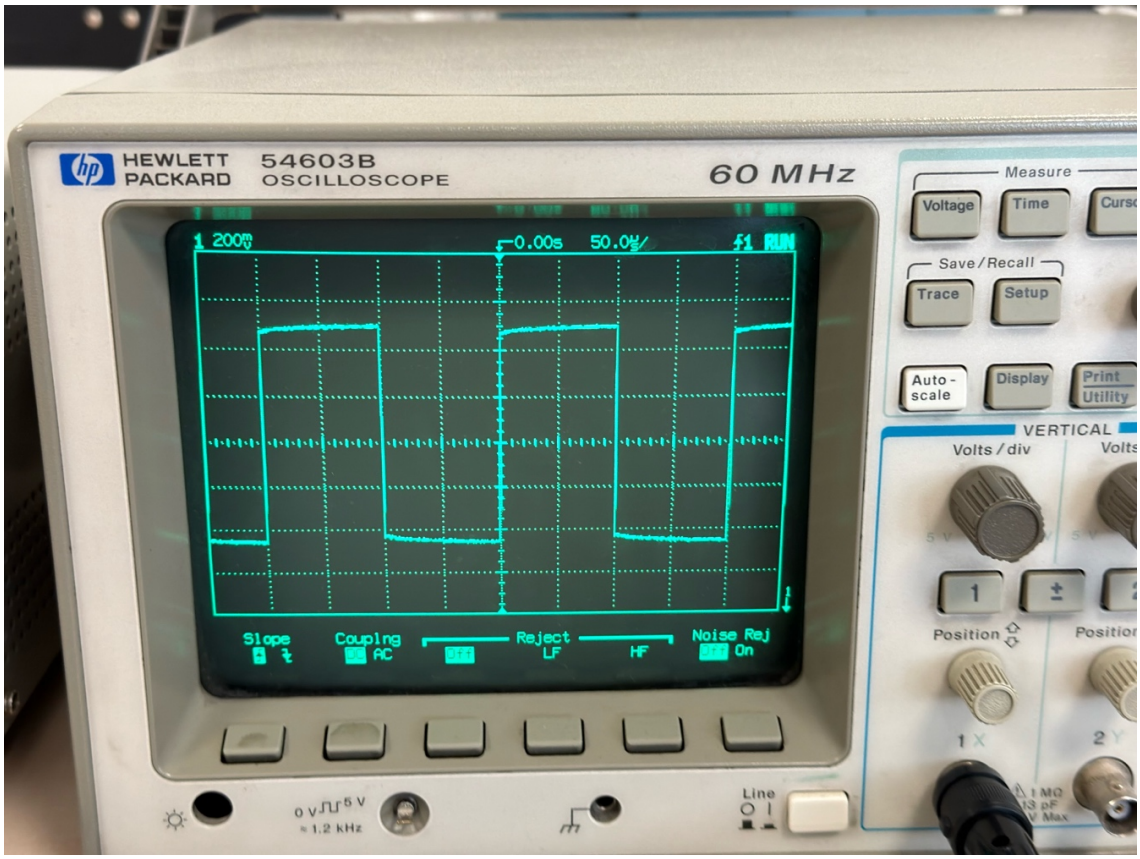


Figure 1.1 Before Compensation,

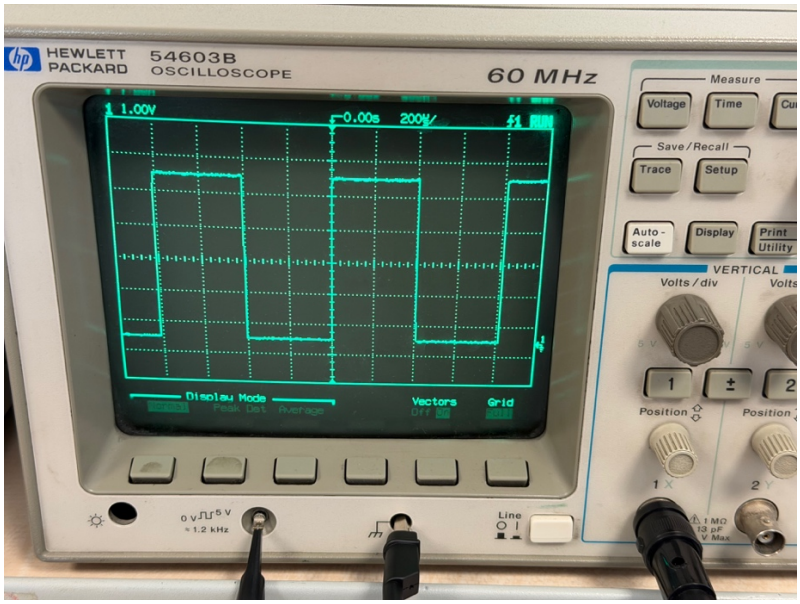


Figure 1.2 After Compensation

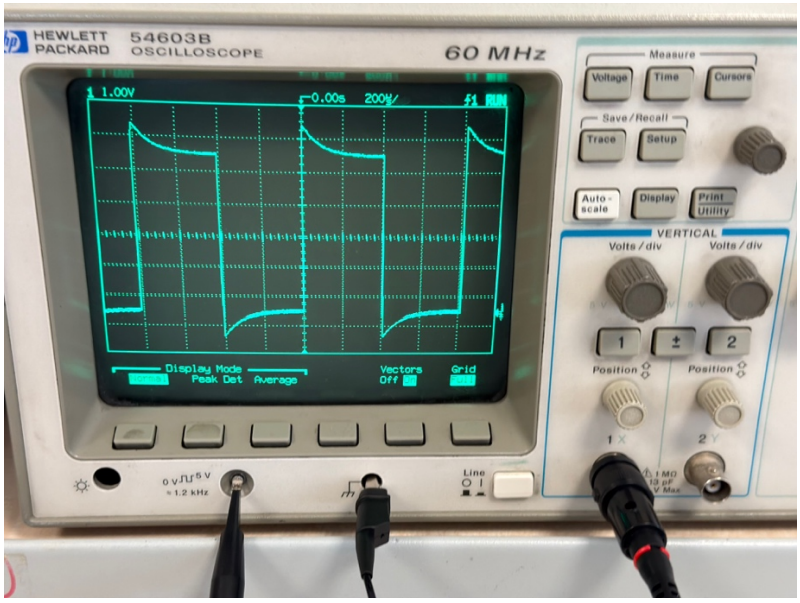


Figure 1.3 Overcompensation

Task 2: Basic Triggering

I used the signal generator to make a 1 kHz sine wave with creating 5 Volt (peak-to-peak). And to cancel de DC component, I set 0 V DC offset on the signal generator. I connected it to the oscilloscope. For the positive edge triggering, the signal was stable and started on the rising slope. On the other hand, negative edge triggering signal started on the falling slope. Initially, rising edge triggering was used, then it was switched to falling edge triggering.

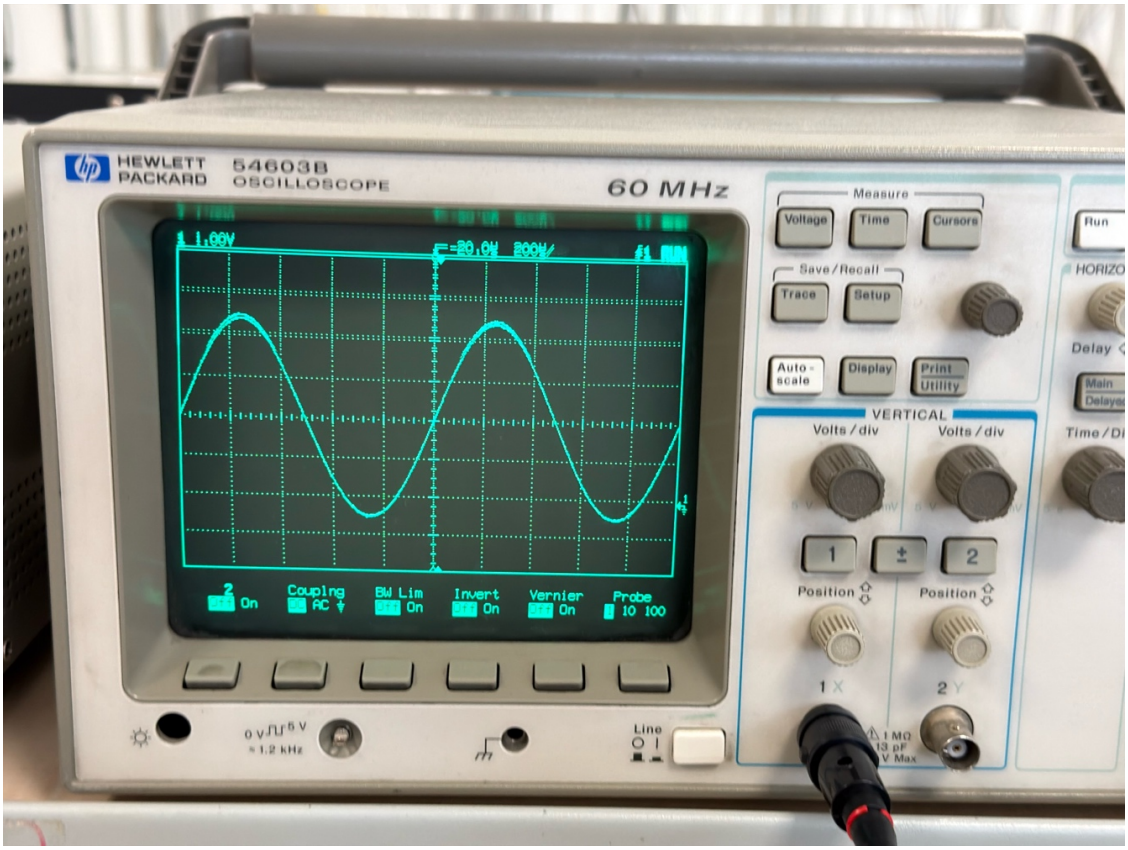


Figure 2.1 5V Pk-Pk 1kHz Signal

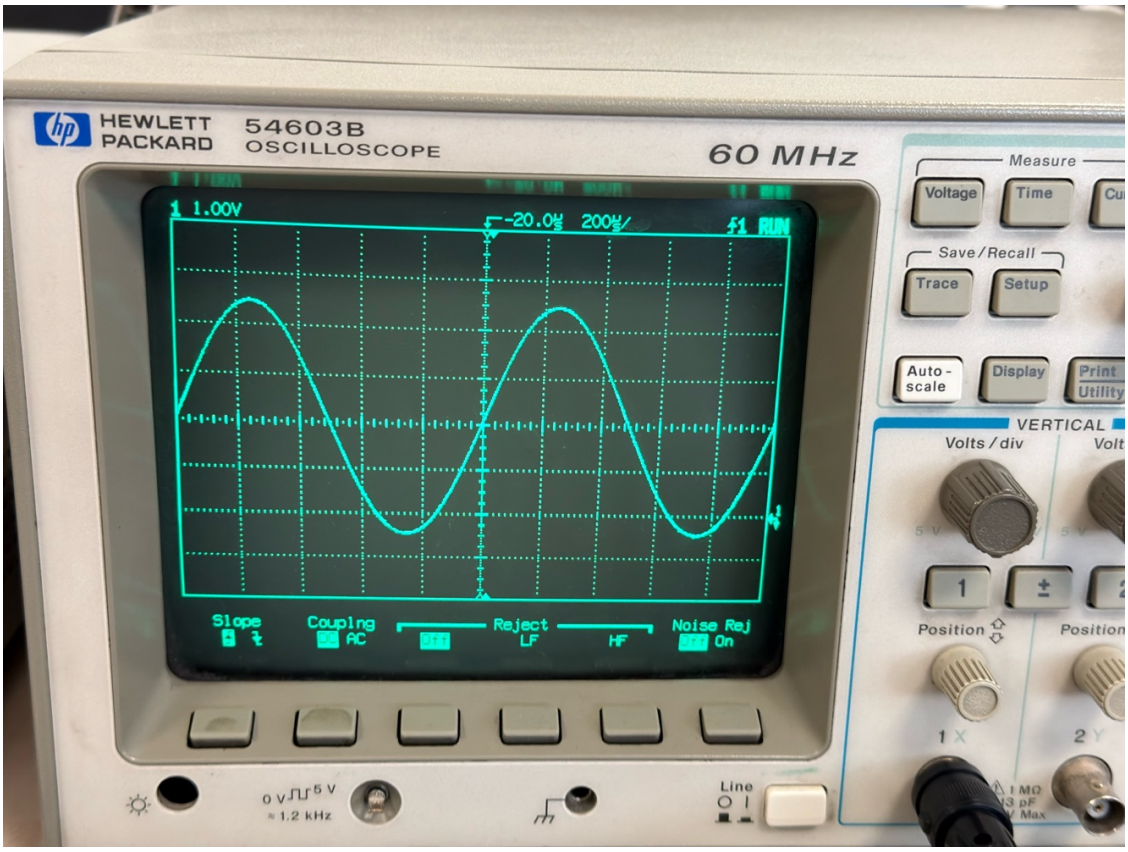


Figure 2.2 "Rising" Slope Trigger

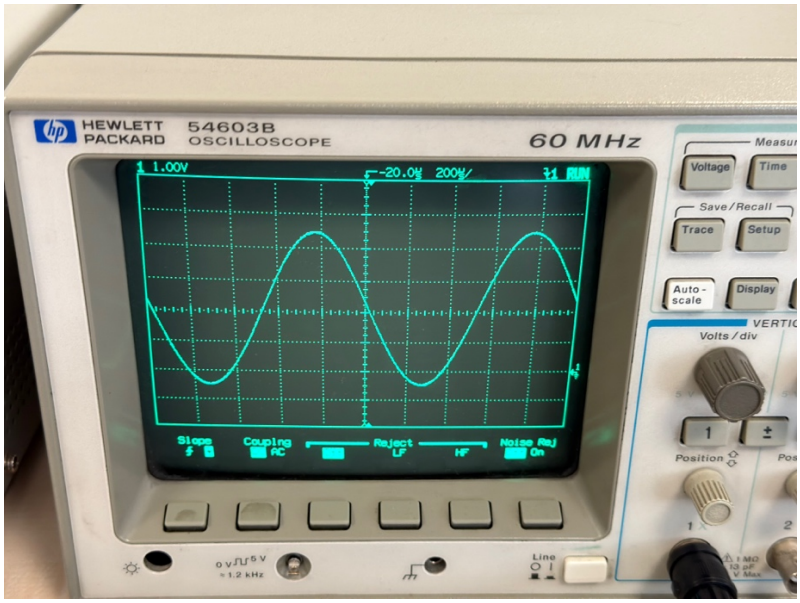


Figure 2.3 "Falling" Slope Trigger

Task 3: Trigger Level

I applied a 1 V (peak-to-peak) triangular wave with 2 kHz frequency to the oscilloscope. The trigger level knob was adjusted manually to observe the stability limits. It was observed that the signal remains stable when the trigger level is set between the voltage range. If the level is set above the peak edge or below the minimum edge, the signal turn into an unstable wave on the screen. Then the oscilloscope failed to synchronize, causing the signal to scroll or flicker.

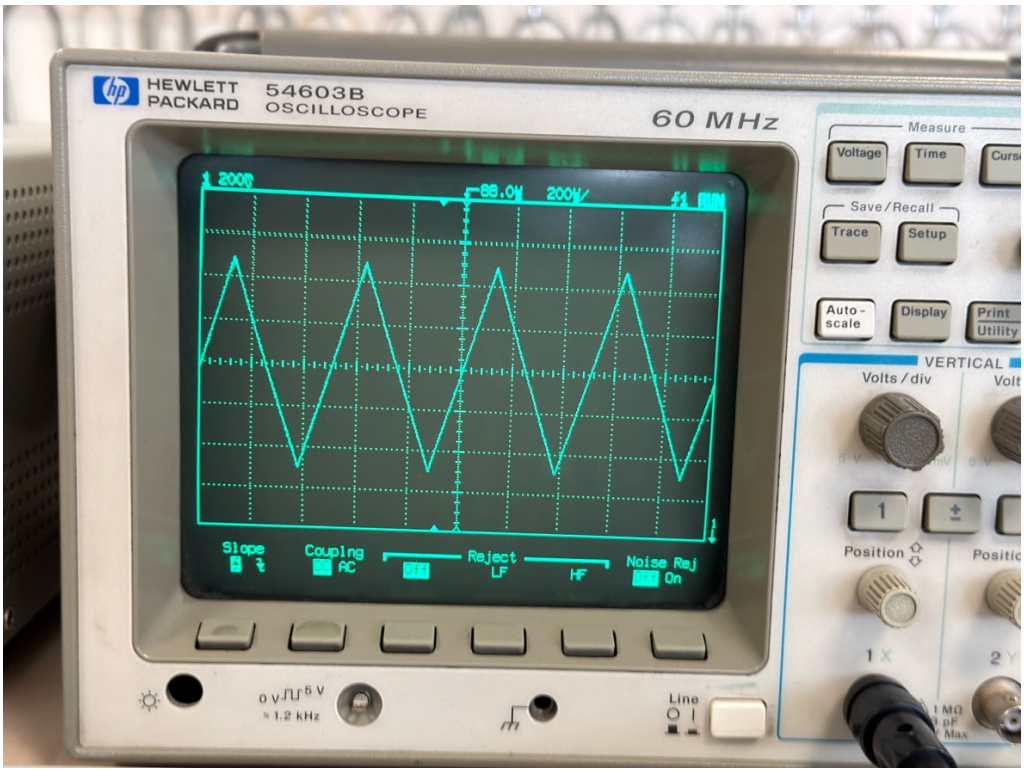


Figure 3.1 1V Pk-Pk 2kHz Stable Signal with Triggering

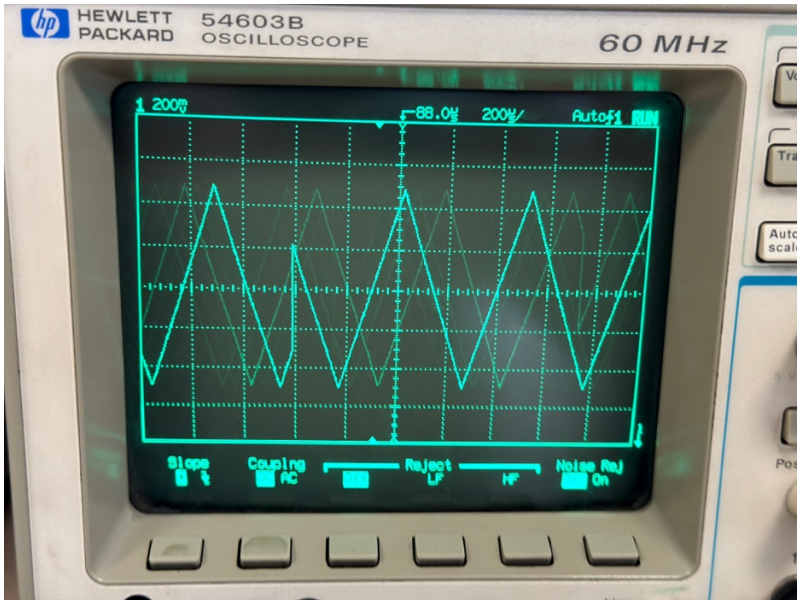


Figure 3.2 1V Pk-Pk 2kHz Unstable Signal without Triggering

Task 4: ADC/DAC and Roll Mode

An Analog-to-Digital (ADC) converts continuous analog signals (Voltage) into digital numbers. This is the type of conversion which oscilloscopes use to display the input signal on the digital screen. A Digital-to-Analog Converter (DAC) converts digital numbers into analog signals which is the opposite of the ADC.

In this part of the experiment, square wave with 1 V (peak-to-peak) and 5kHz frequency is used. Also, I investigate different acquisition modes of the oscilloscope. In “SAMPLE” acquisition mode, oscilloscope only accepts one wave form and send it to display for each period. When operating in “PEAK DETECT” acquisition mode, the oscilloscopes collect two waveforms during each period and send them to screen to display. In “AVERAGE” acquisition mode the oscilloscope takes the average value of different waveforms for each period and displays it on the screen.

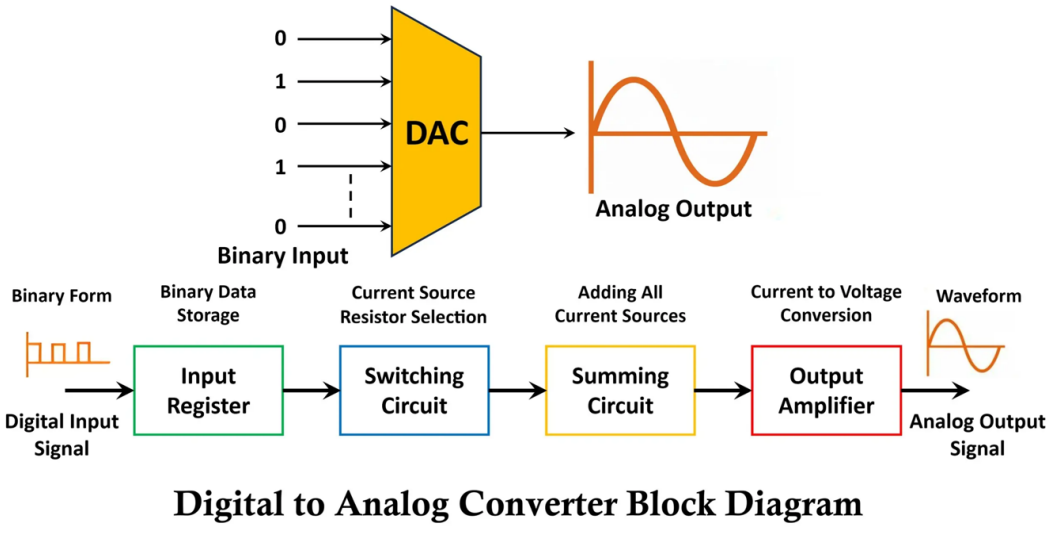


Figure 4.1 Digital to Analog Converter

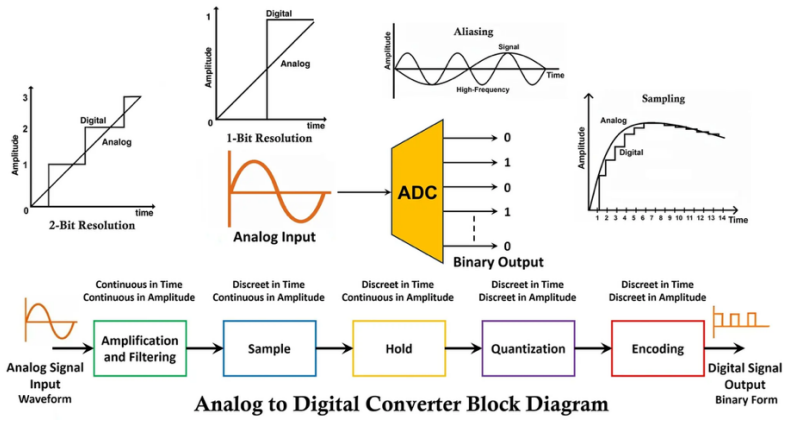


Figure 4.2 Analog to Digital Converter

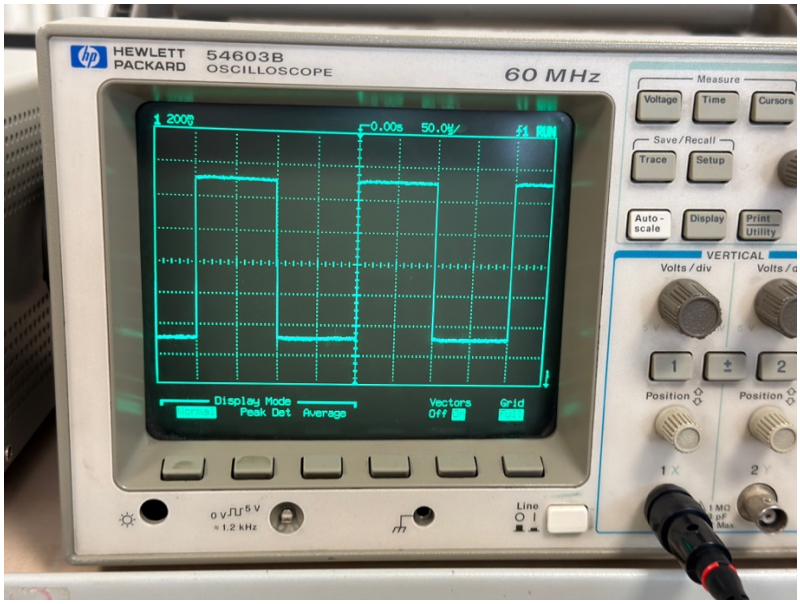


Figure 4.3 1V Pk-Pk 5kHz Square Wave Signal

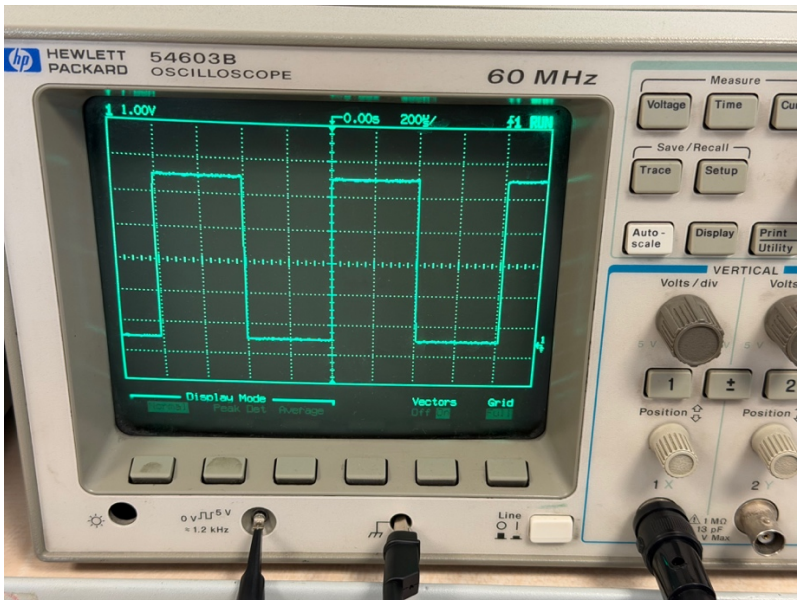


Figure 4.4 Acquired Signal in "SAMPLE" Acquisition Mode

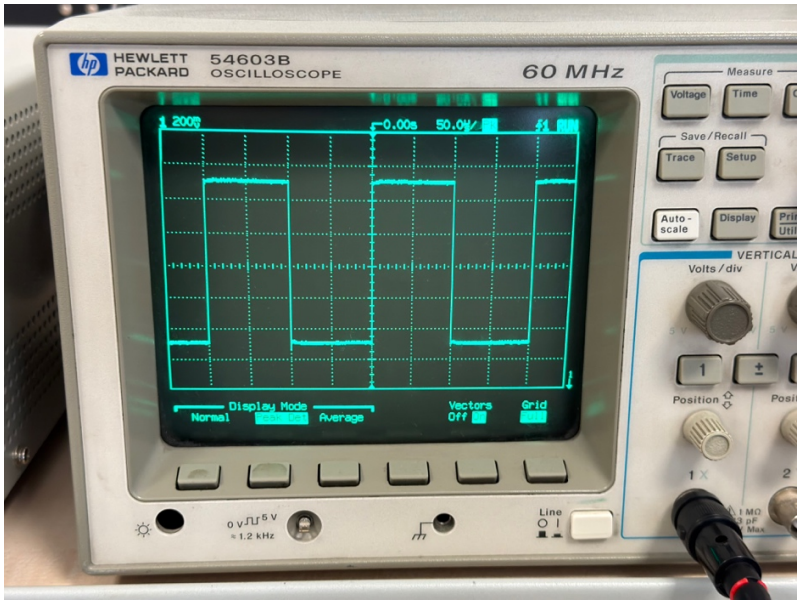


Figure 4.5 Acquired Signal in "PEAK DETECT" Acquisition Mode

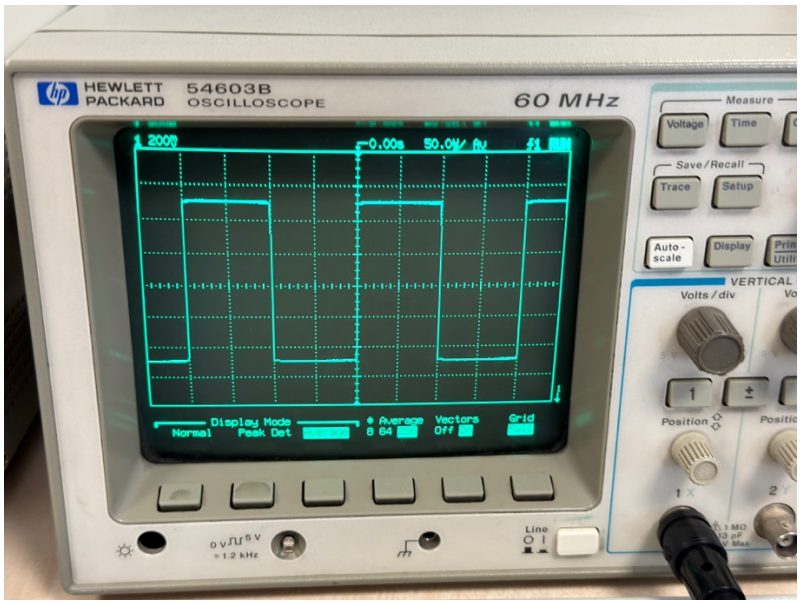


Figure 4.6 Acquired Signal in "Average" Acquisition Mode

Task 5: AC/DC Coupling

I generated a 1kHz square wave with 2V (peak-to-peak) amplitude and a 1V DC offset to observe the coupling effects. First, I used DC coupling, and the oscilloscope allowed both the AC component and the DC offset to pass. The signal appeared shifted up by 1V on the screen. After that I used AC coupling, and the capacitor inside the oscilloscope input blocked the DC component. Therefore, 1V offset was removed and the signal was almost centred around the 0V. This mode is useful for analyzing small AC signals riding on top of large DC voltages.

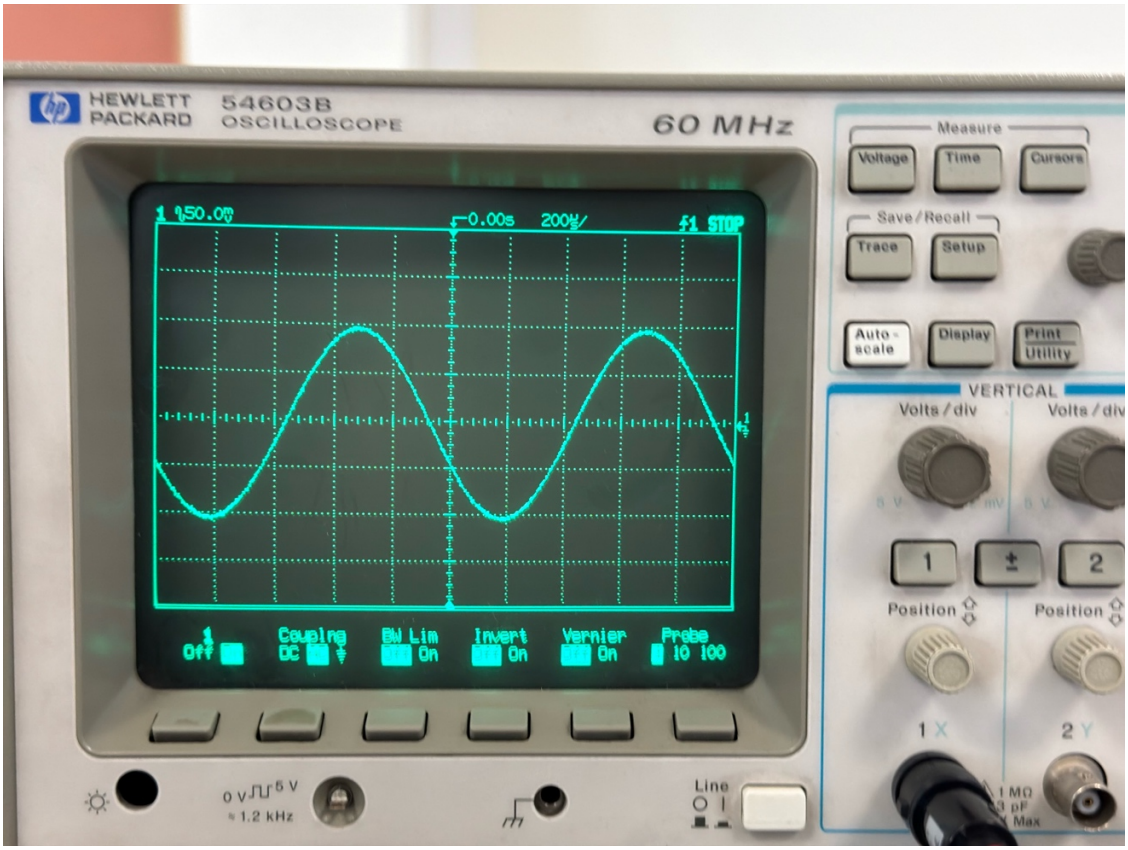


Figure 5.1 2V Pk-Pk 1kHz with 1V offset

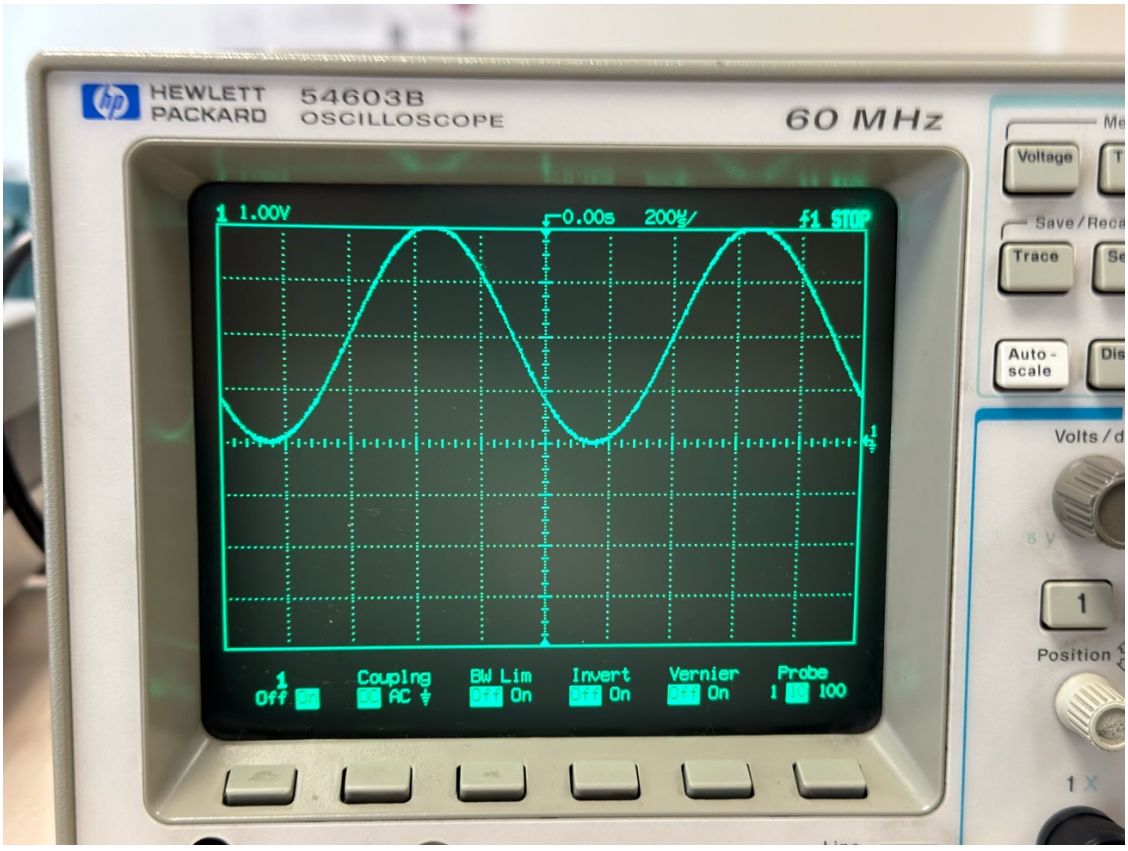


Figure 5.2 DC Coupling

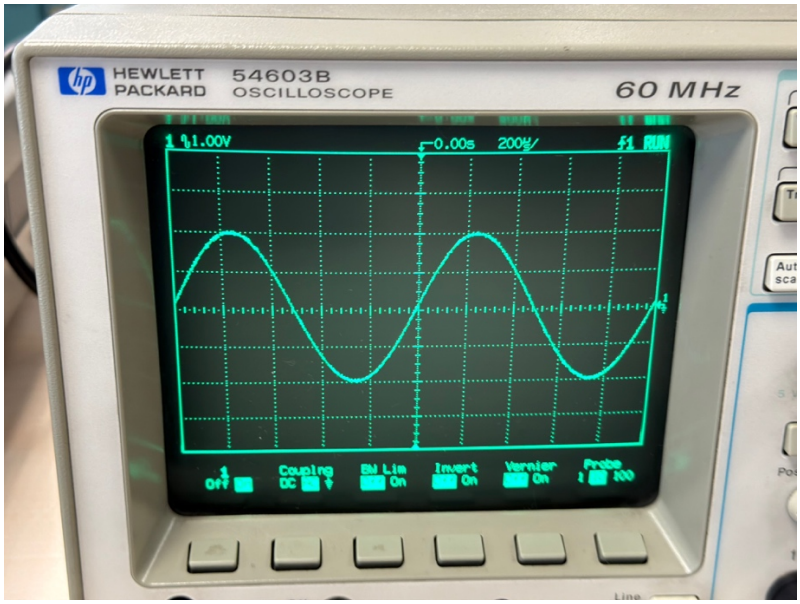


Figure 5.3 AC Coupling

Task 6: Breadboard and Circuit Analysis

I build a simple RC circuit on a breadboard with a $1k\Omega$ resistor and a $1\mu F$ capacitor and a bunch of male-male jumper wires. I used the vertical strips for the connections without soldering and applied a 2V (peak-to-peak) sinusoidal input signal. At 1kHz I displayed the input signal (Channel 1) and the output signal across the capacitor (Channel 2). I observed a slight time delay (Δt) between the peaks of the two sine waves. This shows us that there is a phase shift, where the capacitor voltage lags behind the input due to charging and discharging. At 100 kHz, I increased the frequency significantly. I observed that the output signal (Channel 2) amplitude dropped close to zero. This happens because the capacitor's impedance decreases as frequency increases. At high frequencies, the capacitor behaves almost like a short circuit and sends the signal to ground.

I used the following equation to compute the phase difference between X and Y manually:

$$\varphi = \frac{\Delta t}{T} \times 360^{\circ}$$

1. At 1 kHz Frequency: For the 1 kHz input signal, the period (T) is 1000µs. The time delay (Δt) between the waveforms was measured as approximately 25.2 µs. Using the phase shift formula:

$$\phi = \frac{\Delta t}{T} \times 360^{\circ} = \frac{25.2}{1000} \times 360^{\circ} \approx 9.07^{\circ}$$

This value indicates the phase difference at the lower frequency.

2. At 100 kHz Frequency: When the frequency was increased to 100 kHz, the period (T) became 10μs (10,000 ns). The time delay became extremely small, measured as approximately 2.5 ns. Calculating the phase shift:

$$\phi = \frac{2ns}{10000ns} \times 360^{\circ} = 0.09^{\circ}$$

As observed, at high frequencies, the phase difference approaches zero. These experimental results (9.07° and 0.09°) are consistent with the expected theoretical trend where phase shift decreases significantly as frequency increases.

After setting up the circuit below, the signal generator is being used to create the desired signal.

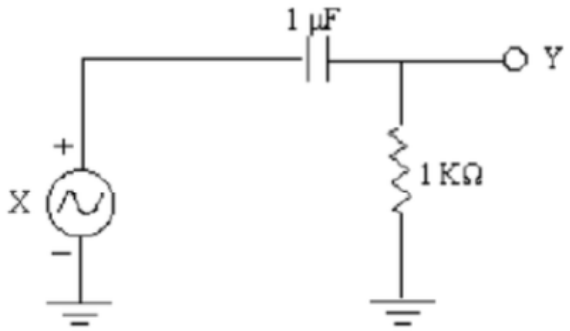


Figure 6.1 Desired Circuit

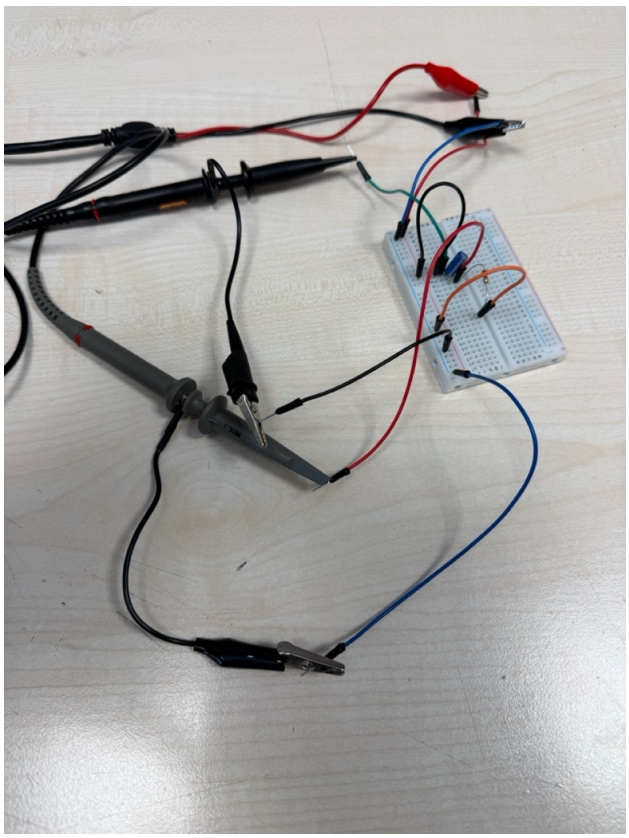


Figure 6.2 RC Circuit on Breadboard

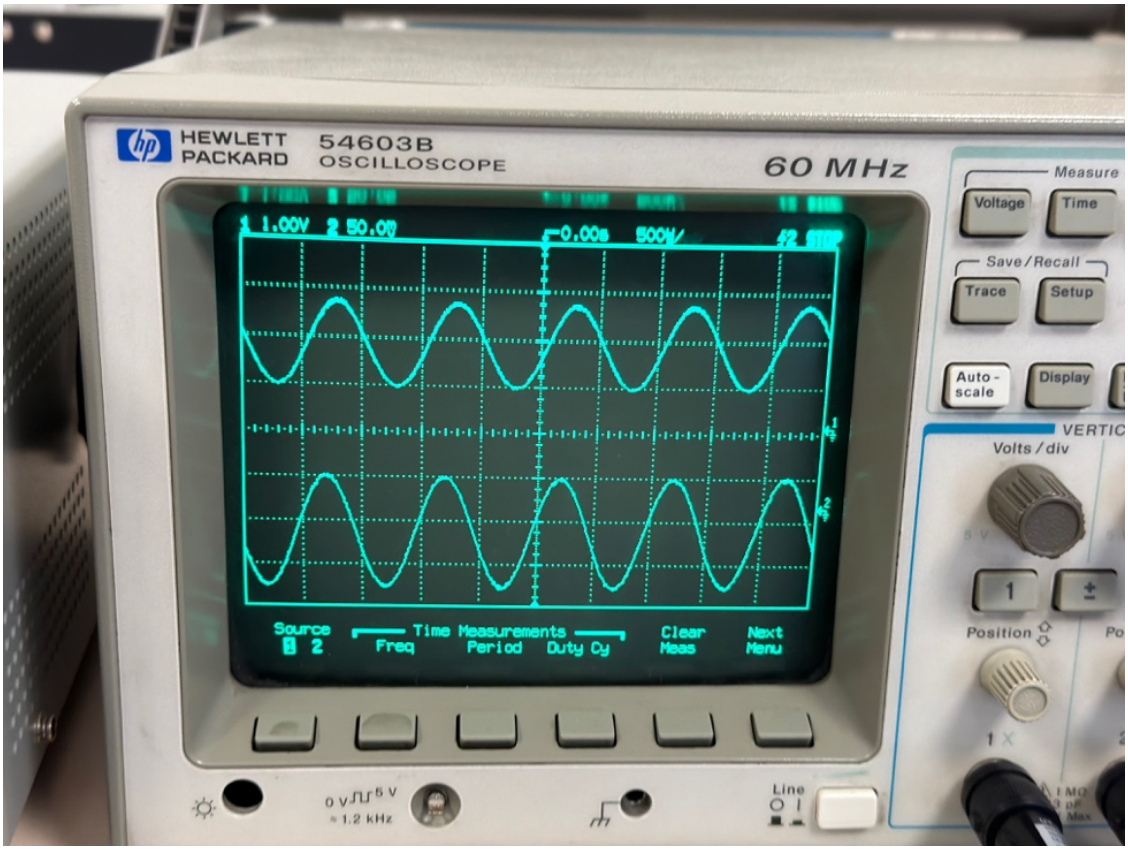


Figure 6.3 2V Pk-Pk 1kHz Signal from Point X and Y

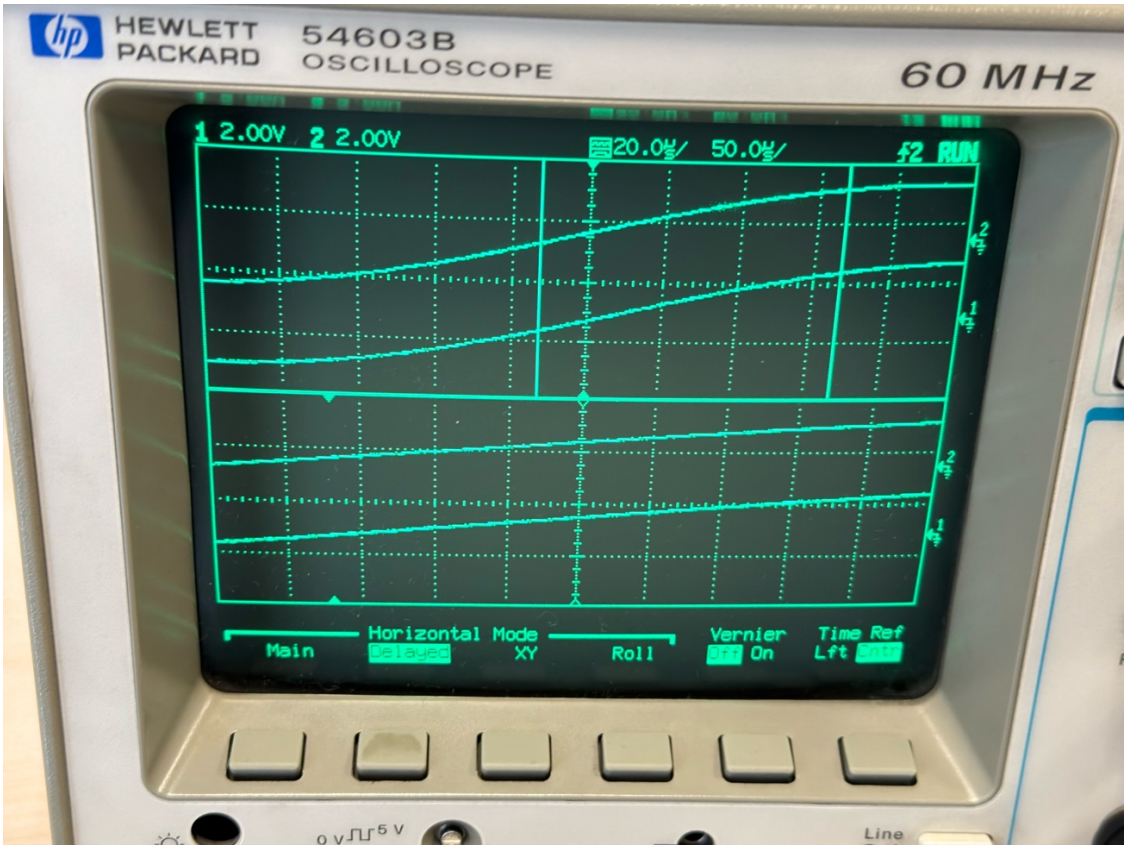


Figure 6.4 Time Difference between Signals from Point X and Y with 1kHz

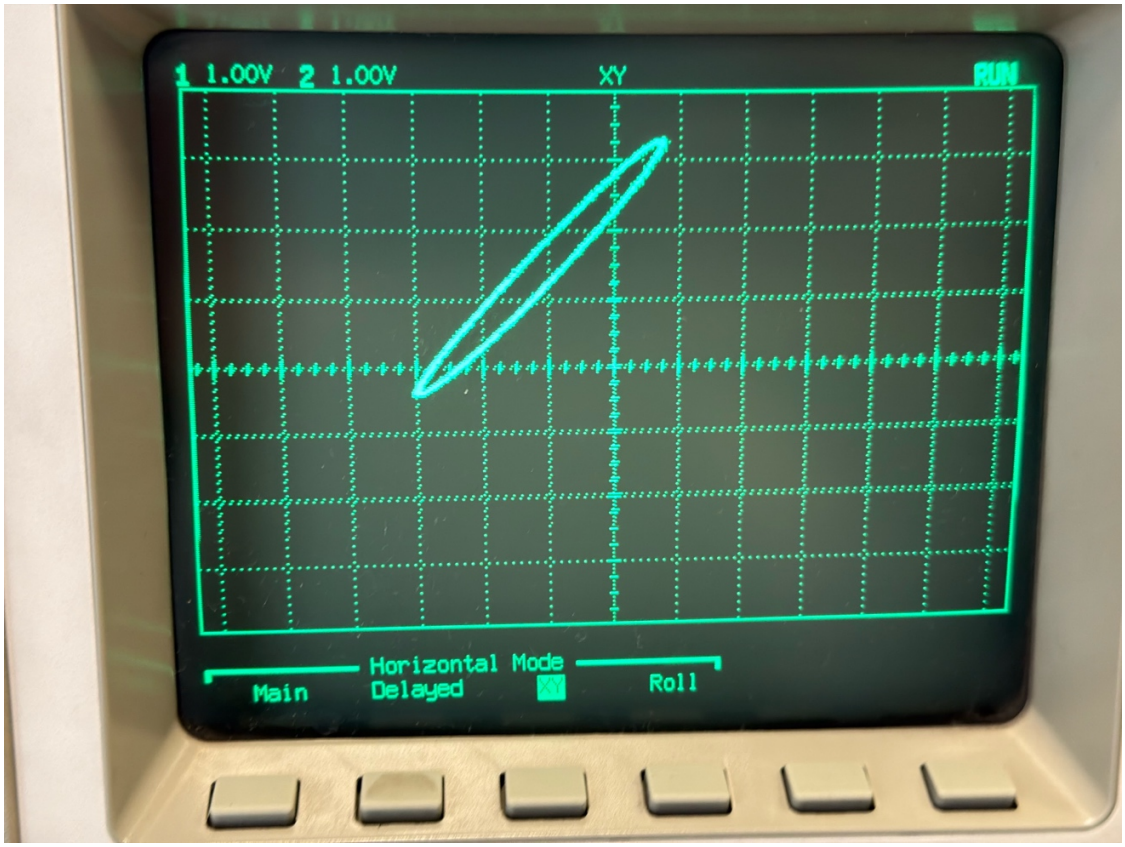


Figure 6.5 The Phase Difference between X and Y

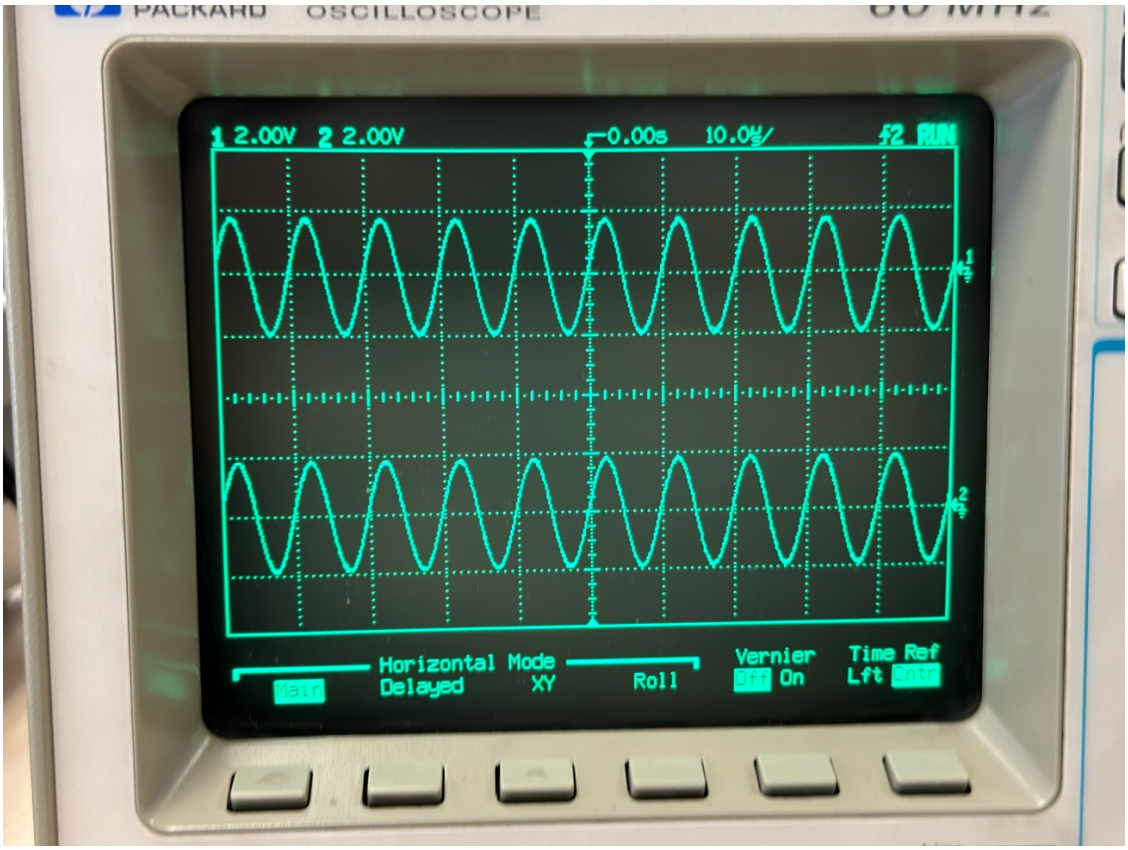


Figure 6.6 2V Pk-Pk 100kHz Signal from Point X to Y

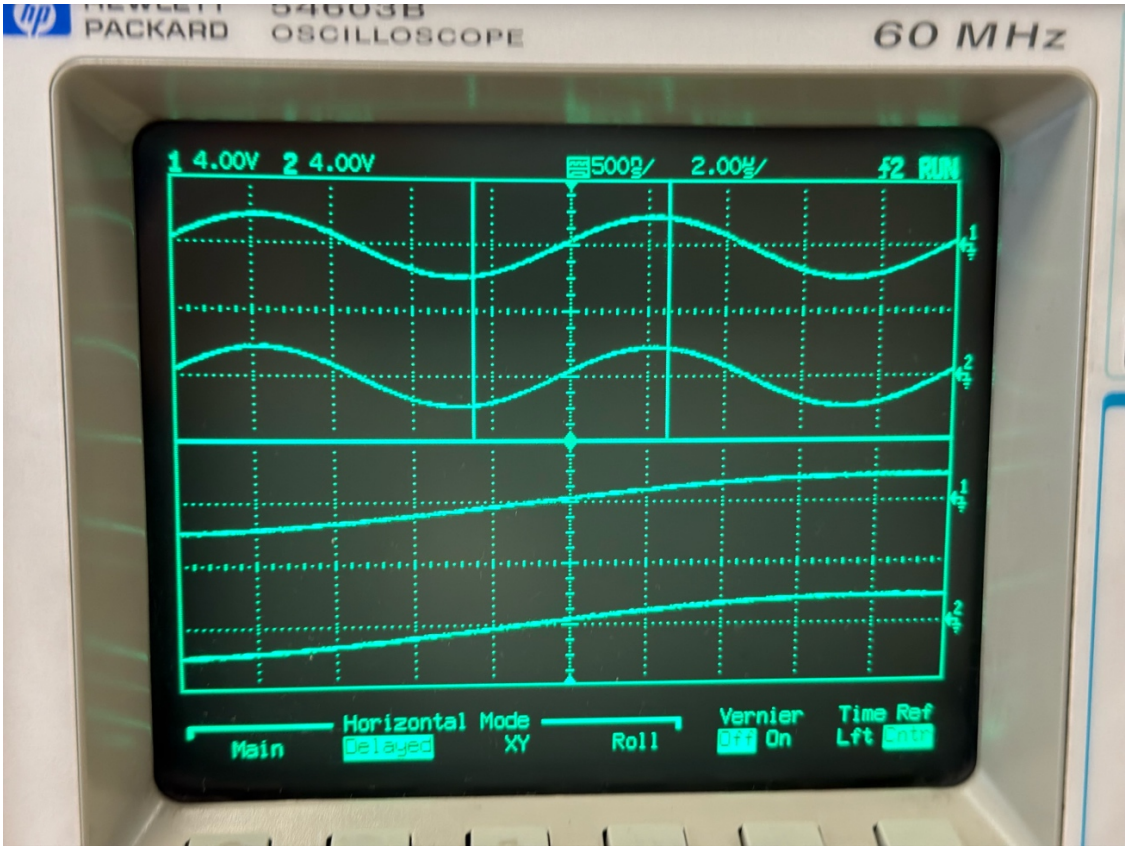


Figure 6.7 The Time Difference between Signals from Point X and Y with 100kHz

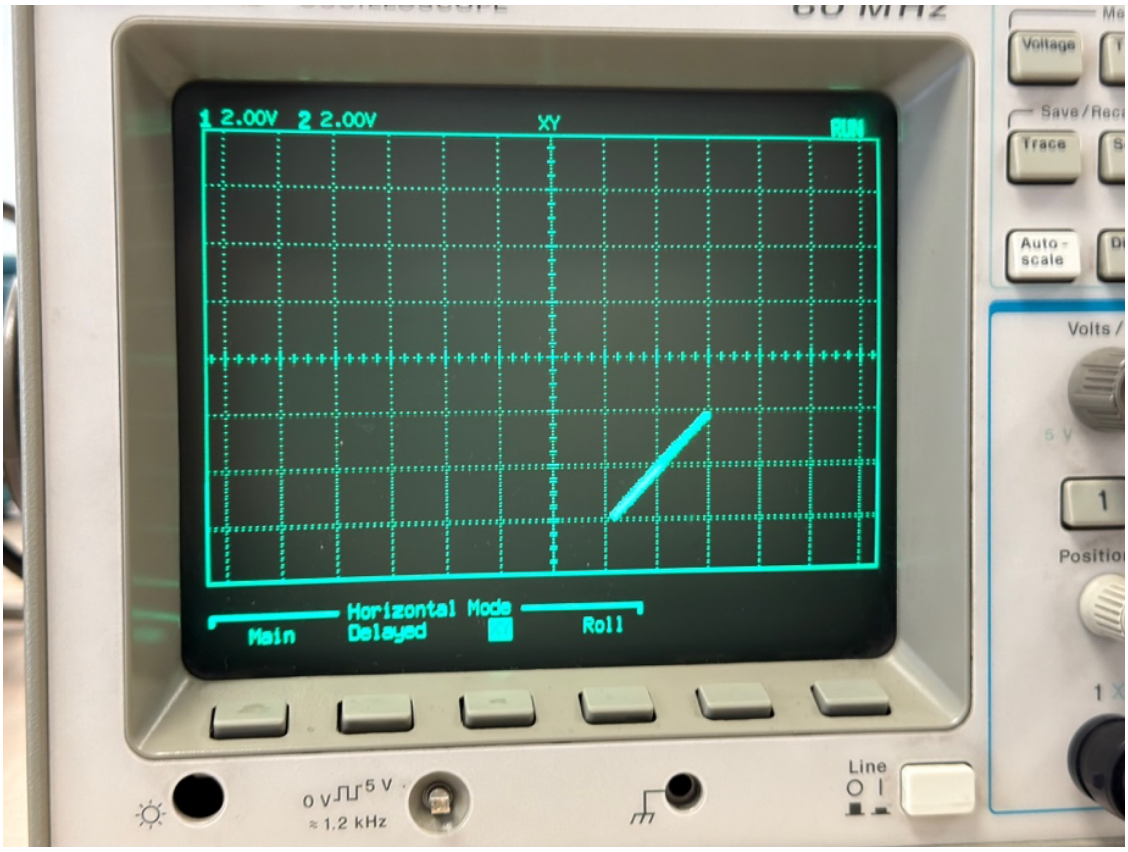


Figure 6.8 The Phase Difference Between X and Y

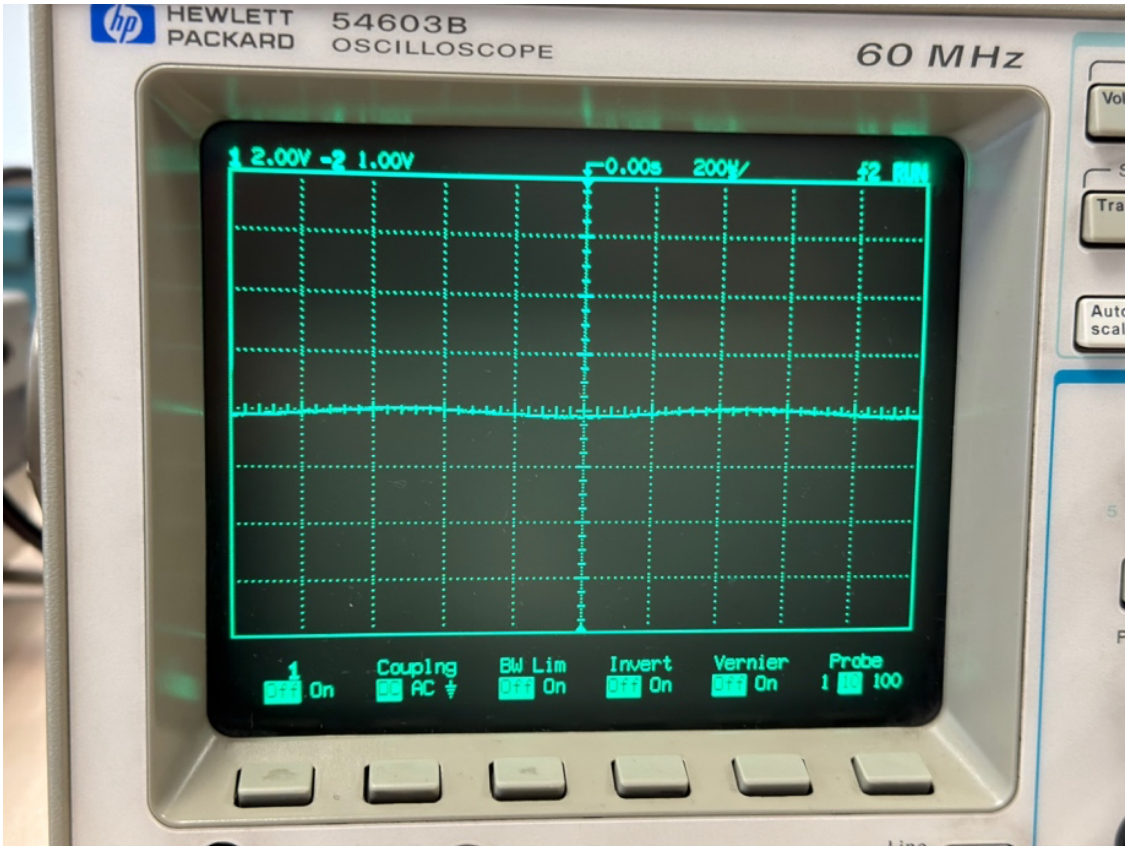


Figure 6.9 Voltage Difference between X and Y

Conclusion:

In this experiment, the fundamental operation of the digital oscilloscope and signal generator was successfully practiced. The necessity of probe compensation for frequency response accuracy was verified. The experiments demonstrated how triggering stabilizes periodic signals. The distinction between AC and DC coupling was observed practically. Finally, the RC circuit experiment provided a clear visualization of how frequency affects both the amplitude and phase (time) delay of a signal, consistent with theoretical circuit analysis.

References:

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